The city reader

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"Towards Sustainable Development"

from *Our Common Future* (1987)

World Commission on Environment and Development
(The Brundtland Commission)

EDITOR'S INTRODUCTION

The physical city is a man-made construct, but the relationship between the city and the surrounding natural environment has always helped to define the character and quality of urban life. If the very first cities were expressions of "hydraulic" civilizations based on the control of water for irrigation, and if the cities of the Industrial Revolution ushered in a new age of unprecedented environmental degradation, it was unpolluted nature – or at least the *idea* of unpolluted nature – that offered a continuing source of intellectual regeneration and moral comparison. The utopian visionaries of the nineteenth and twentieth centuries – Frederick Law Olmsted (p. 364), Ebenezer Howard (p. 371), Le Corbusier (p. 379), and Frank Lloyd Wright (p. 388) – created city plans that balanced the man-made urban areas with parks, public gardens, or surrounding rural zones. And in recent years, the forces of nature ecology, green urbanism, and sustainability have grown in importance and become dominant forces in urban policy and planning.

In 1968, Stanford University biologist Paul Ehlich published *The Population Bomb*, predicting global overcrowding and persistent starvation in the underdeveloped nations by the 1980s. In 1970, the first Earth Day was celebrated in San Francisco. And in 1972, the Club of Rome published its influential report on *The Limits to Growth*, arguing that the world's governments and economic powers needed to begin cutting back on overproduction and overconsumption in the face of uneven development, exploding population growth, and declining resources. All these developments suggested a growing shift toward environmentalism as the new reform paradigm at the very time that capitalism was globalizing and the ideological certainties of the 1960s left were beginning to lose favor. Then, in 1987, came *Our Common Future*, the report of the United Nations-sponsored World Commission on Environment and Development (WCED) and the clarion call for "sustainable development."

The WCED report is commonly called the Brundtland Report, so-named after the chairperson of the Commission, Gro Brundtland of Norway, the only prime minister of a major country to have previously served as environment minister. In its "Call to Action," the Brundtland Report argued that during the twentieth century "the relationship between the human world and the planet that sustains it has undergone a profound change" and that the increased "rate of change is outstripping the ability of . . . our current capabilities." As a result, the world needs to embrace the concept of sustainable development, defined as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs." Such sustainable development, the Report continues, "contains within it . . . the concept of 'needs,' in particular the needs of the world's poor, to which overriding priority should be given."

Some skeptics dismissed the Brundtland Report as just one more attack on free-market capitalism in favor of massive government regulation of the economy. And even some radical environmentalists claimed that "sustainable development" was an oxymoron favored by big business interests to portray capitalism as ecologically benign. But the momentum behind the environmental vision of sustainability sparked by the Brundtland Report continued to grow with the Rio Declaration on Environment and Development of 1992, the Kyoto Protocol on global climate change of 1997–1999, and the World Summit on Sustainable Development in Johannesburg in 2002. Subsequent international climate change summits – and a series of sometimes controversial biennial reports of the UN's Intergovernmental Panel on Climate Change (IPCC) – have proven to be less successful in bringing substantive change in government policies around greenhouse gas emissions, especially when economic crises have redirected policies toward the revival of industrial development to provide jobs for the unemployed. But today, despite continued resistance to the imposition of strict controls on economic development, concerns about climate change and carbon dioxide emissions from the burning of fossil fuels have caused more and more governments to look for new, cleaner sources of energy. And in urban planning, the idea of "sustainability" has been broadly adopted as one of the key concepts behind urban development projects worldwide, especially in Europe and North America and most particularly in the work of New Urbanists (p. 410) and other advocates of environmentally low-impact building techniques, pedestrian-friendly cities, electric automobiles, and renewable energy sources like solar and wind.

For descriptions in this volume of sustainable development applied to urban contexts, consult Timothy Beatley's "Planning for Sustainability in European Cities" (p. 492) and Peter Calthorpe's *Urbanism in the Age of Climate Change* (p. 511). For a more expanded range of views on urban sustainability, consult Stephen Wheeler and Timothy Beatley (eds.), *The Sustainable Urban Development Reader*, 3rd edn (London and New York: Routledge, 2014). The literature on environmentalism in general is huge and ubiquitous, and a notable recent trend has been the perception that dense cities are, *per capita*, the greenest type of human settlement pattern. For example, in "Green Manhattan," David Owen (p. 414) calls for cities to be "more like New York," and in *Triumph of the City* (2011), Edward Glaeser (p. 707) argues that "misguided environmentalism", like building height controls, often pushes urban populations out of cities towards the sprawling suburbs, thereby creating more pollution and more wasteful, unsustainable patterns of land use.

In the end, sustainability is an idea that is most compelling when it cleaves closely to the underlying science of climate change and avoids the exaggeration and sensationalism of journalism and political discourse, but since mass media frenzies and partisan bias are so closely intertwined with environmental policy, many dissenting voices abound, not all of them from "flat earthers" or climate change "deniers." For the best of the questioning views, consult Bjorn Lumborg, *The Skeptical Environmentalist: Measuring the Real State of the World* (Cambridge: Cambridge University Press, 2001) and Stewart Brand, *Whole Earth Discipline* (New York: Penguin, 2010), a remarkably intelligent and challenging manifesto, by one of the founders of the modern environmental movement, for a new approach to sustainability that embraces formerly taboo subjects like nuclear power and genetically modified crops.

A CALL FOR ACTION

Over the course of this century, the relationship between the human world and the planet that sustains it has undergone a profound change.

When the century began, neither human numbers nor technology had the power radically to alter planetary systems. As the century closes, not only do vastly increased human numbers and their activities have that power, but major, unintended changes are occurring in the atmosphere, in soils, in waters, among plants and animals, and in the relationships among all of these. The rate of change is outstripping the ability of scientific disciplines and our current capabilities to assess and advise. It is frustrating the attempts of political and economic institutions, which evolved in a different, more fragmented world, to adapt and cope. It deeply worries many people who are seeking ways to place those concerns on the political agendas.

The onus lies with no one group of nations. Developing countries face the obvious life-threatening challenges of desertification, deforestation, and pollution, and endure most of the poverty associated with environmental degradation. The entire human family of nations would suffer from the disappearance of rain forests in the tropics, the loss of plant and animal species, and changes in rainfall patterns. Industrial nations face the life-threatening challenges of toxic chemicals, toxic wastes, and acidification. All nations may suffer from the releases by industrialized countries of carbon dioxide and of gases that react with the ozone layer, and from any future war fought with the nuclear arsenals controlled by those nations. All nations will have a role to play in changing trends, and in righting an international economic system that increases rather than decreases inequality, that increases rather than decreases numbers of poor and hungry.

The next few decades are crucial. The time has come to break out of past patterns. Attempts to maintain social and ecological stability through old approaches to development and environmental protection will increase instability. Security must be sought through change. The Commission has noted a number of actions that must be taken to reduce risks to survival and to put future development on paths that are sustainable. Yet we are aware that such a reorientation on a continuing basis is simply beyond the reach of present decision-making structures and institutional arrangements, both national and international.

This Commission has been careful to base our recommendations on the realities of present institutions, on what can and must be accomplished today. But to keep options open for future generations, the present generation must begin now, and begin together.

To achieve the needed changes, we believe that an active follow-up of this report is imperative. It is with this in mind that we call for the UN General Assembly, upon due consideration, to transform this report into a UN Programme on Sustainable Development. Special follow-up conferences could be initiated at the regional level. Within an appropriate period after the presentation of this report to the General Assembly, an international conference could be convened to review progress made, and to promote follow-up arrangements that will be needed to set benchmarks and to maintain human progress.

First and foremost, this Commission has been concerned with people – of all countries and all walks of life. And it is to people that we address our report. The changes in human attitudes that we call for depend on a vast campaign of education, debate, and public participation. This campaign must start now if sustainable human progress is to be achieved.

The members of the World Commission on Environment and Development came from 21 very different nations. In our discussions, we disagreed often on details and priorities. But despite our widely differing backgrounds and varying national and international responsibilities, we were able to agree to the lines along which change must be drawn.

We are unanimous in our conviction that security, wellbeing, and very survival of the planet depend on such changes, now.

A THREATENED FUTURE

The Earth is one but the world is not. We depend on one biosphere for sustaining our lives. Yet each community, each country, strives for survival and prosperity with little regard for its impact on others. Some consume the Earth's resources at a rate that would leave little for future generations. Others, many more in number, consume far too little and live with the prospect of hunger, squalor, disease, and early death.

Yet progress has been made. Throughout much of the world, children born today can expect to live longer and be better educated than their parents. In many parts, the newborn can also expect to attain a higher standard of living in a wider sense. Such progress provides hope as we contemplate the improvements still needed, and also as we face our failures to make this Earth a safer and sounder home for us and for those who are to come.

The failures that we need to correct arise both from poverty and from the short-sighted way in which we have often pursued prosperity. Many parts of the world are caught in a vicious downwards spiral: Poor people are forced to overuse environmental resources to survive from day to day, and their impoverishment of their environment further impoverishes them, making their survival ever more difficult and uncertain. The prosperity attained in some parts of the world is often precarious, as it has been secured through farming, forestry, and industrial practices that bring profit and progress only over the short term.

Societies have faced such pressures in the past and, as many desolate ruins remind us, sometimes succumbed to them. But generally these pressures were local. Today the scale of our interventions in nature is increasing and the physical effects of our decisions spill across national frontiers. The growth in economic interaction between nations amplifies the wider consequences of national decisions. Economics and ecology bind us in ever-tightening networks.

Today, many regions face risks of irreversible damage to the human environment that threaten the basis for human progress.

These deepening interconnections are the central justification for the establishment of this Commission. We traveled the world for nearly three years, listening. At special public hearings organized by the Commission, we heard from government leaders, scientists, and experts, from citizens' groups concerned about a wide range of environment and development issues, and from thousands of individuals – farmers, shanty-town residents, young people, industrialists, and indigenous and tribal peoples.

We found everywhere deep public concern for the environment, concern that has led not just to protests but often to changed behaviour. The challenge is to ensure that these new values are more adequately reflected in the principles and operations of political and economic structures.

We also found grounds for hope: that people can cooperate to build a future that is more prosperous, more just, and more secure; that a new era of economic growth can be attained, one based on policies that sustain and expand the Earth's resource base; and that the progress that some have known over the last century can be experienced by all in the years ahead. But for this to happen, we must understand better the symptoms of stress that confront us, we must identify the causes, and we must design new approaches to managing environmental resources and to sustaining human development.

SYMPTOMS AND CAUSES

Environmental stress has often been seen as the result of the growing demand on scarce resources and the pollution generated by the rising living standards of the relatively affluent. But poverty itself pollutes the environment, creating environmental stress in a different way. Those who are poor and hungry will often destroy their immediate environment in order to survive: They will cut down forests, their livestock will overgraze grasslands; they will overuse marginal land; and in growing numbers they will crowd into congested cities. The cumulative effect of these changes is so far-reaching as to make poverty itself a major global scourge.

On the other hand, where economic growth has led to improvements in living standards, it has sometimes been achieved in ways that are globally damaging in the longer term. Much of the improvement in the past has been based on the use of increasing amounts of raw materials, energy, chemicals, and synthetics and on the creation of pollution that is not adequately accounted for in figuring the costs of production processes. These trends have had unforeseen effects on the environment. Thus today's environmental challenges arise both from the lack of development and from the unintended consequences of some forms of economic growth.

[. . .]

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It contains within it two key concepts:

- the concept of 'needs', in particular the essential needs of the world's poor, to which overriding priority should be given; and
- the idea of limitations imposed by the state of technology and social organization on the environment's ability to meet present and future needs.

Thus the goals of economic and social development must be defined in terms of sustainability in all countries – developed or developing, market-oriented or centrally planned. Interpretations will vary, but must share certain general features and must flow from a consensus on the basic concept of sustainable development and on a broad strategic framework for achieving it. Development involves a progressive transformation of economy and society. A development path that is sustainable in a physical sense could theoretically be pursued even in a rigid social and political setting. But physical sustainability cannot be secured unless development policies pay attention to such considerations as changes in access to resources and in the distribution of costs and benefits. Even the narrow notion of physical sustainability implies a concern for social equity between generations, a concern that must logically be extended to equity within each generation.

THE CONCEPT OF SUSTAINABLE DEVELOPMENT

The satisfaction of human needs and aspirations is the major objective of development. The essential needs of vast numbers of people in developing countries

– for food, clothing, shelter, jobs – are not being met, and beyond their basic needs these people have legitimate aspirations for an improved quality of life. A world in which poverty and inequity are endemic will always be prone to ecological and other crises. Sustainable development requires meeting the basic needs of all and extending to all the opportunity to satisfy their aspirations for a better life.

Living standards that go beyond the basic minimum are sustainable only if consumption standards everywhere have regard for long-term sustainability. Yet many of us live beyond the world's ecological means, for instance in our patterns of energy use. Perceived needs are socially and culturally determined, and sustainable development requires the promotion of values that encourage consumption standards that are within the bounds of the ecologically possible and to which all can reasonably aspire.

Meeting essential needs depends in part on achieving full growth potential, and sustainable development clearly requires economic growth in places where such needs are not being met. Elsewhere, it can be consistent with economic growth, provided the content of growth reflects the broad principles of sustainability and non-exploitation of others. But growth by itself is not enough. High levels of productive activity and widespread poverty can coexist, and can endanger the environment. Hence sustainable development requires that societies meet human needs both by increasing productive potential and by ensuring equitable opportunities for all. An expansion in numbers can increase the pressure on resources and slow the rise in living standards in areas where deprivation is widespread. Though the issue is not merely one of population size but of the distribution of resources, sustainable development can only be pursued if demographic developments are in harmony with the changing productive potential of the ecosystem.

A society may in many ways compromise its ability to meet the essential needs of its people in the future – by overexploiting resources, for example. The direction of technological developments may solve some immediate problems but lead to even greater ones. Large sections of the population may be marginalized by ill-considered development.

Settled agriculture, the diversion of watercourses, the extraction of minerals, the emission of heat and noxious gases into the atmosphere, commercial forests, and genetic manipulation are all examples of human intervention in natural systems during the course of development. Until recently, such interventions were small in scale and their impact limited. Today's interventions are more drastic in scale and impact, and more threatening to life-support systems both locally and globally. This need not happen. At a minimum, sustainable development must not endanger the natural systems that support life on Earth: the atmosphere, the waters, the soils, and the living beings.

Growth has no set limits in terms of population or resource use beyond which lies ecological disaster. Different limits hold for the use of energy, materials, water, and land. Many of these will manifest themselves in the form of rising costs and diminishing returns, rather than in the form of any sudden loss of a resource base. The accumulation of knowledge and the development of technology can enhance the carrying capacity of the resource base. But ultimate limits there are, and sustainability requires that long before these are reached, the world must ensure equitable access to the constrained resource and reorient technological efforts to relieve the pressure.

Economic growth and development obviously involve changes in the physical ecosystem. Every ecosystem everywhere cannot be preserved intact. A forest may be depleted in one part of a water-shed and extended elsewhere, which is not a bad thing if the exploitation has been planned and the effects on soil erosion rates, water regimes, and genetic losses have been taken into account. In general, renewable resources like forests and fish stocks need not be depleted provided the rate of use is within the limits of regeneration and natural growth. But most renewable resources are part of a complex and interlinked ecosystem, and maximum sustainable yield must be defined after taking into account system-wide effects of exploitation.

As for nonrenewable resources, like fossil fuels and minerals, their use reduces the stock available for future generations. But this does not mean that such resources should not be used. In general the rate of depletion should take into account the criticality of that resource, the availability of technologies for minimizing depletion, and the likelihood of substitutes being available. Thus land should not be degraded beyond reasonable recovery. With minerals and fossil fuels, the rate of depletion and the emphasis on recycling and economy of use should be calibrated to ensure that the resource does not run out before acceptable substitutes are available. Sustainable development requires that the rate of depletion of nonrenewable resources should foreclose as few future options as possible.

Development tends to simplify ecosystems and to reduce their diversity of species. And species, once extinct, are not renewable. The loss of plant and animal species can greatly limit the options of future generations; so sustainable development requires the conservation of plant and animal species.

So-called free goods like air and water are also resources. The raw materials and energy of production processes are only partly converted to useful products. The rest comes out as wastes. Sustainable development requires that the adverse impacts on the quality of air, water, and other natural elements are minimized so as to sustain the ecosystem's overall integrity.

In essence, sustainable development is a process of change in which the exploitation of resources, the direction of investments, the orientation of technological development, and institutional change are all in harmony and enhance both current and future potential to meet human needs and aspirations.

FIVE

"Urbanism in the Age of Climate Change"

Peter Calthorpe

EDITORS' INTRODUCTION

In this selection, Peter Calthorpe – a noted architect, urban designer, author and leader of the New Urbanism movement and the Congress of the New Urbanism (p. 410) – addresses what many consider the greatest challenge humanity has ever faced – global climate change. Calthorpe takes it as a given that climate change is an imminent threat and potential catastrophe. The science is overwhelming that human settlements and disproportionately the urban agglomerations where more than half the world's population now live are causing global warming to a significant degree, though scientific investigations are proceeding and debates raging on how much change is occurring and where. On a *per capita* basis very dense cities like New York produce less carbon and damage the environment less than more spread out areas as David Owen describes (p. 414). Because emission or releasing carbon into the atmosphere is the biggest culprit, planning carbon-neutral cities for the future and remediating impacts of damage already done to the atmosphere by carbon emissions since cities like Manchester began pouring coal smoke into the atmosphere during the Industrial Revolution as Engels describes (p. 53) are a major urban planning challenge everywhere in the world. Calthorpe provides grim information on the harm we have already done and a sobering assessment of how difficult it will be to reduce carbon emissions enough to assure the long term viability of cities. For the US, what Calthorpe calls "the twelve percent solution" – cutting per capita greenhouse gas emissions to just 12 percent of their current level by 2050 – will require deep changes, not only in our energy sources, technology, and conservation means, but also in urban design, culture and lifestyles. Thus a theme running through Calthorpe's selections is the importance of an interdisciplinary holistic approach that combines insights from the natural and social science to devise workable solutions.

Calthorpe's unique contribution to the debate about what to do about climate change – one he has been developing for his entire professional career – is the role that urbanism can play in reducing global climate change. Developing solar, wind, wave, geothermal, biomass, perhaps even nuclear power in place of oil and coal will be critical. So will conservation and restoration of habitat that can absorb carbon. So will alternative energy vehicles and better stewardship of the environment and natural resources. But for Calthorpe a big contribution to reducing global climate change is better planning of cities and metropolitan regions – particularly more compact cities with better linkage between land use and transportation. That is exactly the type of design Calthorpe has been working on all his professional life. As an architect, urban designer, and author, Calthorpe began thinking about how to design cities in ways that conserve natural resources, protect the environment, and use alternative energy sources and incorporating them into plans and developments long before world opinion caught up to the challenge. Calthorpe's New Urbanist designs are at the forefront of designs for compact, livable, mixed-use, pedestrian-friendly cities. A cause for optimism is that he argues city designs that reduce carbon emissions can also achieve many of these same livability goals.

The Brundtland Commission's 1987 ten-year window in which they argued that worldwide development patterns would have to be reversed (p. 404) has long since closed. Yet virtually every country is still pursuing destructive development that will require programs that are much more difficult, costly, and disruptive than they would have been if leaders heeded the Brundtland Commission's warning a quarter of a century ago. No matter how effectively urban planners change plans for cities of the future and no matter how much money and effort governments devote to the problem, so much damage has now been done that cities everywhere in the world will experience severe climate change-related problems. The scientific consensus is that temperatures will likely rise by about 2° Celsius (3.6° Fahrenheit) by mid-century. Changes of this magnitude will have enormous impacts, though exactly what the impacts will be and where they will occur is still being studied and hotly debated. Heatwaves will likely increase mortality among people and animals. Storm surges and sea level rise will require costly flood protection systems and may flood cities built near sea level regardless. Water scarcity will become a problem as mountain snow packs and glaciers melt. Global climate change is likely to produce excessive rainfall in some areas of the world and drought in other areas, affecting agriculture and food availability. It will have complex effects on ecosystems. These changes may require the relocation of millions of people and in hard-hit areas may produce political instability and even provoke wars. If the more dire scenarios occur, cities will have to severely restrict auto use, build costly seawalls and levies, create new reservoirs and water transmission lines, evacuate low-lying areas, retrofit buildings at risk from more and more frequent and violent hurricanes and tornados, and build resilient cities as Vale describes (p. 618).

Greenhouse gases seep into the atmosphere from sources as varied as smokestacks, automobile tailpipes, livestock manure, and air-conditioning equipment. They come from transportation, electricity and heat used in buildings, industry, agriculture, and landfills and many other sources. Accordingly, any effort to reduce global warming must consider many sources and a new and extremely holistic way of thinking about how to develop cities.

How did we get into this dreadful situation? In addition to our wasteful and unsustainable use of resources documented by the Brundtland Commission (p. 404), the enormous and exponential growth of the world's population described by Kingsley Davis (p. 19) and the emergence of enormous megacity regions described by Tingwei Zhang (p. 687) help explain our predicament. There are many more people now than in the recent past and more than half of them now live in cities. Kenneth T. Jackson's description of American's love affair with automobiles (p. 73) contributes to the explanation. More and more people everywhere in the world participate in the auto-centered culture Jackson describes, depleting oil and contributing to GHG emissions. Automobile use is exploding in China, India, Brazil, and other populous countries with large cities that – together with the US – contribute the most to global climate change. The low density, sprawling land use patterns that Robert Bruegmann (p. 218) and Shlomo Angel (p. 537) describe increase auto dependency.

At the level of urban planning practice, planning to address global climate change is a complex task that requires holistic and interdisciplinary approaches connecting the insights of biologists and transit planners, agronomists, hydrologists, economists, and many other disciplines and professional fields.

There is movement towards addressing climate change at many levels. At the national and multinational scales, important – albeit insufficient – multinational agreements have been made. Some regions, such as the European Union, and many of the world's countries have set goals for reducing GHG emissions and are pursuing strategies to meet those goals.

At regional and subnational scales, many governments and agencies have developed plans and policies for reducing GHG emissions. A majority of US states now have some sort of climate change plan. Metropolitan regional agencies are also adapting to the new reality. Many are reworking plans for public transit systems to promote transit-oriented development and ensure more compact urban form.

Calthorpe notes that cities are inherently green and big cities greener than small cities. On a *per capita* basis they require less land, less auto travel, and use less energy and emit less carbon than other settlement types. He points out that residents of New York City – America's largest city – emit, on average, only one third as much carbon as US residents overall – a point nicely developed in David Owen's selection "Green Manhattan" (p. 414).

A growing number of cities, towns, and rural communities have adopted climate change plans and policies to reduce GHG emissions. Some require public buildings to be certified under the US Green Building Council's LEED (Leadership in Energy and Environmental Development) rating system or encourage better insulated passive houses that use little or no off-site energy. Other local approaches include investing in bicycle and pedestrian transportation systems, creating district heating and cooling systems, promoting small-scale community energy systems using wind and solar power, reducing the land area covered by dark asphalt (which absorbs heat) and painting roofs light colors (which reflects heat).

Urban solutions to global climate change need to take place at every scale from the self-conscious individual household all the way to multinational organizations. But the key to urbanism as a climate change strategy is regional planning and concerted metropolitan-wide action. Calthorpe argues that whole systems design functions best at the regional scale.

Nongovernmental organizations and citizens' groups are also doing a great deal to educate the public and demonstrate practical solutions to reduce global climate change. Civic engagement and social capital independent of government as Robert D. Putnam describes (p. 154) are essential parts of the solution to global climate change.

While global climate change is likely to be the greatest challenge this generation of urban planners face, Calthorpe argues that it is also an opportunity – a chance to create far more livable, equitable, and sustainable communities and lifestyles. Climate change may finally provide the impetus for the kind of urbanism he has long advocated: livable cities that are human-scale, compact, walkable, pedestrian-friendly, transit-oriented, and that harmonize the built and natural environment.

Peter Calthorpe is the founder of Calthorpe Associates an urban design firm based in Berkeley, California. An author, practicing architect, and one of the founders of the Congress for the New Urbanism, his writings and innovative projects have had a large impact worldwide.

This selection is from *Urbanism in the Age of Climate Change* (Washington, DC: Island Press, 2013). Other of Calthorpe's books include *The Next American Metropolis* (New York: Princeton Architectural Press, 1993), *The Regional City: Planning for the End of Sprawl*, co-authored with William Fulton (Washington, DC: Island Press, 2001), and *Sustainable Communities*, co-authored with Sym Van der Ryn (San Francisco: Sierra Club Books, 1986).

Other recent book on cities and global climate change include Harriet Burkeley, *Cities and Climate Change* (London and New York: Routledge, 2013) and Stephen Wheeler's *Climate Change and Social Ecology* (London and New York: Routledge. 2012).

A seminal book on global climate change is *Our Choice: A Plan to Solve the Climate Crisis* (Emmaus, PA: Rodale, 2009) by former US vice-president and Nobel Peace Prize winner Al Gore. The most authoritative books describing the science of global climate change are the US National Research Council, *Climate Change: Evidence, Impacts, and Choices* (Washington, DC: National Research Council, 2013) and the United Nations Intergovernmental Panel on Climate Change (IPCC), *Climate Change 2013: The Physical Science Basis* (Geneva: IPCC, 2007).

Other books on global climate change include Simon Foxell, *A Carbon Primer for the Built Environment* (London and New York: Routledge, 2014), Arnold Bloom, *Global Climate Change* (Basingstoke: Sinauer, 2008), Andrew Dessler and Edward A. Parson, *The Science and Politics of Global Climate Change* (Cambridge: Cambridge University Press, 2006), Tim Flannery, *The Weather Makers: How Man is Changing the Climate and What It Means for Life on Earth* (New York: Grove, 2005), Diane Dumanoski, *The End of the Long Summer: Why We Must Remake Our Civilization to Survive on a Volatile Earth* (New York: Crown Publishers, 2009), Mark Lynas, *Six Degrees: Our Future on a Hotter Planet* (Washington, DC: National Geographic, 2008), Elizabeth Kolbert, *Field Notes from a Catastrophe: Man, Nature, and Climate Change* (New York: Bloomsbury, 2006), Karen McGlothlin, *Global Climate Change* (Lanham, MD: Roman and Littlefield, 2006), Thomas R. Karl, Jerry M. Melillo, Thomas C. Peterson, and Susan J. Hassol, *Global Climate Change Impacts in the United States* (Cambridge: Cambridge University Press, 2009), George Monbiot, *Heat: How to Stop the Planet From Burning* (Cambridge, MA: South End Press, 2007), Fred Pearce, *With Speed and Violence: Why Scientists Fear Tipping Points in Climate Change* (Boston: Beacon, 2006), Joseph Romm, *Hell and High Water: Global Warming – the Solution and the Politics – and What We Should Do* (New York: Morrow, 2007), Michael E Schlesinger et al. (eds.), *Human Induced Climate Change: An Interdisciplinary Assessment* (Cambridge: Cambridge University Press, 2007).

For contrarian views see Patrick J. Michaels and Robert C. Balling, Jr., *Climate of Extremes: Global Warming Science They Don't Want You to Know* (Washington, DC: Cato Institute, 2010) and Bjorn Lomborg, *The Skeptical Environmentalist: Measuring the Real State of the World* (Cambridge: Cambridge University Press, 2001).

For a critique of climate change skeptics see James Hoggan and Richard Littlemore, *Climate Change Coverup: The Crusade to Deny Global Warming* (Petersburg, VA: Graystone, 2009).

INTRODUCTION

* * *

[U]rbanism is often missing from the proposed remedies for climate change, job growth, and environmental stress: it is the invisible wedge in the pie chart of green solutions.

. . . I define the term urbanism broadly—by qualities, not quantities: by intensity, not density; by connectivity, not just location. Urbanism is always made from places that are mixed in uses, walkable, human scaled, and diverse in population: that balance cars with transit; that reinforce local history; that are adaptable; and that support a rich public life. Urbanism can come in many forms, scales, locations, and densities. Many of our traditional villages, streetcar suburbs, country towns, and historic cities are "urban" by this definition. Urbanism often resides beyond our downtowns.

While urbanism will vary by geography, culture, and economy, traditional urbanism always manifests the vitality, complexity, and intimacy that defined our finest cities and towns for centuries. By this definition suburbs can be "urban" if they are walkable and mixed use, and cities can easily be the reverse—just visit any central city "urban renewal" district. Traditional urbanism is not just a central city location, a historic district, a downtown, or a "phase" we passed over; it is an evolving platform for our most essential needs—and at this moment in history it is fundamental to shaping a sustainable future.

The solution to the climate change and energy challenge does not necessarily pit suburb versus city; rather it requires their reintegration into sustainable regional forms . . . [B]oth must co-evolve into more integrated forms, establishing a seamless interface.

Certainly cities are green. On a per capita basis, they require less land, less auto travel, and less energy and they emit less carbon. But this message may well oversimplify the complex multilayered urban and regional strategies that are key to our future. More than stand-alone "sustainable communities" or even "green cities," we now need "sustainable regions"—places that carefully blend a broad range of technologies, settlement patterns, and lifestyles. Only a regional plan can create a framework for communities of differing scales and intensities, for transportation choices that can significantly offset auto dependence, and for environmental systems and green technologies that function at both the large and small scales. Whole systems design functions best at the regional scale.

Unfortunately urbanism so defined has been on the wane for the last half century. Our cities and towns have been on a high-carbon diet—and our metropolitan regions have become, in short, obese. Oil is like a high-sugar and high-starch diet for cities; it expands the waistline without nourishing strength or resilience. Urban neighborhoods are like healthy diets: they build on unique places and local history, they use natural ingredients and mix them well, they tend toward local sources, and they are lean. America's postwar suburbs are like fast food: their history and sense of place trumped by mass production: their ingredients dominated by a few generic staples; their resources distant and large; and their infrastructure highly subsidized. Our urban footprint—its physical size and resource demands—has expanded in unsustainable ways for too long.

As a remedy, this book will advance the following propositions. First, that urbanism—compact and walkable development—will arise naturally if the built-in bias of our current infrastructure investments, financial structures, zoning norms, and public policies is reformed. Second, that such urbanism, when mixed with simple conservation technologies, can have a major impact in reducing carbon emissions and energy demand. Third, that urbanism is the most cost-effective solution to climate change, more so than most renewable technologies. And finally, that urbanism's many collateral benefits—economic, social, and environmental—enhance its desirability and economics. In short, urbanism is the foundation for a low-carbon future and is our least-cost option.

This book specifically focuses on the United States' unique opportunities and challenges regarding climate change. Since 1850, the United States has contributed close to 30 percent of the globe's cumulative carbon emissions—more than any other country and more than the entire European Union. We represent a disproportionate share of the problem and therefore have a special obligation for leadership and change. Moreover, a U.S. solution would demonstrate a low-carbon future married to middle-class prosperity, a model of a sustainable future that affords both economic development and environmental frugality.

Too often we see this challenge only in technical terms, within the domain of industrial efficiencies, new power generation sources, or green technologies. Instead, I will attempt to paint a picture of a future that sees climate change and energy through the lens of lifestyles, land use, urbanism, and, most significantly, design of the metropolitan region.

But it is not just the threat of climate change or the depletion of energy resources that will dramatically redirect our patterns of settlement. The lines of pressure are converging from many directions: limits of environmentally rich land and clean water are being felt throughout the country; shifts in family size and workforce are changing our social structure; issues of environmental and personal health are mounting; costs of capital and time are reordering investments; and, not least, a new search for identity, community, and a sense of place is motivating many peoples' lives. It is my thesis that a future that responds to all of these pressures will also best address the climate change crisis.

In fact, these wide-ranging environmental, social and economic challenges should not, and realistically cannot, be resolved individually. I have always been suspicious of single-issue causes—no matter how worthy—mostly because they are often blind to both unintended consequences and important collateral benefits. Urbanism's effects reverberate well beyond carbon emissions, and that is exactly why it can become such a powerful solution to the climate change challenge: it is propelled by many other needs. The economics of urbanism reach from simple infrastructure and energy savings to public health, affordable housing, and land conservation. In addition, it involves more qualitative outcomes that relate to social capital, economic equity, and quality of life. Perhaps the most important contribution of this book will be to quantify many of the co-benefits that complement the carbon reductions of a more sustainable urban form.

* * *

The great recession of 2008, and its underlying real estate meltdown, was more than just a crisis of credit structures and banking policies. It was a manifestation of a deeper reality: that many of our communities and lifestyles are unsustainable—too auto dependent, too land intensive, too isolated, and, in the end, too expensive to own and operate. Our development patterns became as toxic as the financial structures that underwrote them. In plain fact, our land use patterns were, and still remain, precariously out of sync with our most profound economic, social, and environmental needs.

URBANISM AND CLIMATE CHANGE

I take as a given that climate change is an imminent threat and potentially catastrophic—the science is now clear that we are day by day contributing to our demise. In addition, I believe that an increase in fuel costs due to declining oil reserves is also inevitable. The combination of these two global threats presents an economic and environmental challenge of unparalleled proportions—and, lacking a response, the potential for dire consequences. These challenges will in turn bring into urgent focus the way our buildings, towns, cities, and regions shape our lives and our environmental footprint. Beyond a transition to clean energy sources, I believe that urbanism—compact, diverse, and walkable communities—will play a central role in addressing these twin threats. In fact, responding to climate change and our coming energy challenge without a more sustainable form of urbanism will be impossible.

Many deny either the timing or the reality of these challenges. They argue that global demand for oil will not outstrip production and that climate change is overstated, nonexistent, or somehow not related to our actions. Setting aside such debates, this book accepts the premise that both climate change and peak oil are pressing realities that need aggressive solutions.

The two challenges are deeply linked. The science tells us that if we are to arrest climate change, our goal for carbon emissions should be just 20 percent of our 1990 level by 2050. That, combined with a projected U.S. population increase of 130 million people means each person in 2050 would need to be emitting on average just 12 percent of his or her current greenhouse gases (GHG)—what I will call here the "12% Solution." If we can achieve the 12% Solution to offset climate change, we will simultaneously reduce our fossil-fuel dependence and demonstrate a sustainable model of prosperity. Such a low-carbon future will inherently reduce oil demands at rates that will allow a smoother transition to alternative fuels—and the next economy.

In addition to these twin environmental challenges, the United States has two other systemic forces to reckon with in the next generation: an aging population and a more diverse middle class with less wealth. We are now a country in which a third of the population are baby boomers or older and less than a quarter are traditional families with kids. And for the past decade, median income has actually fallen; in fact, the typical American household saw its inflation-adjusted income decline by more than $2,000 between 1999 and 2008. So, at the same time that we must respond to climate change and rising energy costs, we must also adjust our housing stock to fit a changing demographic and find a more frugal form of prosperity.

Such a transformation will require deep change, not just in energy sources, technology, and conservation measures but also in urban design, culture, and lifestyles, More than just deploying green technologies and adjusting our thermostats, it will involve rethinking the way we live and the underlying form of our communities. The good news is that our environmental, social, and economic challenges have a shared solution in urbanism. Shaping regions that reduce oil dependence simultaneously reduces carbon emissions, costs less for the average household, and creates healthy, integrated places for our seniors: one solution for multiple challenges.

The urban solution involves both technology and design. For example, we will need to dramatically reduce the number of miles we drive as well as develop less carbon intensive vehicles. It will mean living and working in buildings that demand significantly less energy as well as powering them with renewable sources. It will involve the kinds of food we eat, the kinds of homes we build, the ways travel and the kinds of communities we inhabit. It will certainly involve giving up the idea of any single "silver bullet" solution (whether solar or nuclear, conservation or carbon capture, adaptation or mitigation) and understanding that such a transformation will involve all of the above—and, perhaps most important, that they are all interdependent.

In fact, the viability of new technologies and clean energy sources will depend on the success of our conservation efforts at the regional, community, and building scales, which in turn will be determined by our basic lifestyles and the urban forms that support our changing demographics, The key will be designing the right mix of strategies, a "whole systems" rather than a "checklist" approach to climate change, energy, and economics.

There are three interdependent approaches to these nested challenges: lifestyle, conservation, and clean energy, Lifestyle involves hew we live—the way we get around, the size of our homes, the foods we cat, and the quantity of goods we consume. These depend in turn on the type of communities we build and the culture we inhabit—degrees of urbanism. Conservation revolves around technical efficiencies—in our buildings, cars, appliances, utilities, and industrial systems—as well as preserving the natural resources that support us all, our global forests, ocean ecologies, and farmlands. These conservation measures are simple, they save money, and they are possible now. The third fix, clean energy, is what we have been most focused on: new technologies for solar, wind, wave, geothermal, biomass, and even a new generation of nuclear power or fusion. These energy sources are sexy, they are relatively expensive, and they will be available sometime soon. All three approaches will be essential, but here I focus on the first two—lifestyle and conservation—because they are, in the end, our most cost effective and easily available tools.

The intersection of lifestyle and conservation is urbanism. Consider that in the United States industry represents 29 percent of our GHG emissions; agriculture and other non-energy-related activities, just 9 percent; and freight and planes, another 9 percent. This 47 percent total represents the GHG emissions of the products we buy, the food we eat, the embodied energy of all our possessions, and all the shipping involved in getting them to us. The remaining 53 percent depends on the nature of our buildings and personal transportation system—the realm of urbanism. As a result, urbanism, along with a simple combination of transit and more efficient buildings and cars, can deliver much of our needed GHG reductions.

Perhaps just as important as greenhouse gas reductions and oil savings is the fact that urbanism generates a fortuitous web of co-benefits—it is our most potent weapon against climate change because it does so much more. Urbanism's compact forms lead to less land consumed and more farmland, parks, habitat, and open space preserved. A smaller urban footprint results in less development costs and fewer miles of roads, utilities and services to build and maintain, which then leads to fewer impervious surfaces. less polluted storm runoff, and more water directed back into aquifers.

More compact development leads to lower housing costs as lower land and infrastructure costs affect sales prices and taxes. Urban development means a different mix of housing types—fewer large single-family lots; more bungalows and townhomes—but in the end provides more housing choices for a more diverse population. It means less private space but more shared community places—more efficient and less expensive overall. Urbanism is more suited to an aging population, for whom driving and yard maintenance are a growing burden, and for working families seeking lower utility bills and less time spent commuting.

Urbanism leads to fewer miles driven, which then leads to less gas consumed and less dependence on foreign oil supplies, less air pollution, less carbon emissions. Fewer miles also leads to less congestion, lower emissions, lower road construction and maintenance costs, and fewer auto accidents. This then leads to lower health costs because of fewer accidents and cleaner air, which is reinforced by more walking, bicycling, and exercising, which in turn contributes to lower obesity rates. And more walking leads to more people on the streets, safer neighborhoods, and perhaps stronger communities.

The feedback loops go on. More urban development means more compact buildings—less energy needed to heat and cool, lower utility bills, less irrigation water, and, once again, less carbon in the atmosphere. This then leads to lower demands on electric utilities and fewer new power plants, which again results in less carbon and fewer costs. . . .

But for the past fifty years, our economy and society have been operating on the premise that "more is more" and "bigger is better": bigger homes, bigger yards, bigger cars with bigger engines, bigger budgets, bigger institutions, and, finally, bigger energy sources. In contrast, urbanism naturally tends toward a "small is beautiful" philosophy. This then involves trade-offs: less private space but perhaps a richer public realm; less private security but perhaps a safer community; less auto mobility but more convenient transit. Compact development does mean smaller yards, fewer cars, and less private space for some. On the other hand, it can dramatically reduce everyday costs and leave more time for family and community. The question is not which is right and which is wrong or that it must be all one way or the other—urban ism works best with blends. The question is how such trade-offs fit with our emerging demographics, our desires, our needs, our economic means—and perhaps our sense of what a good life really is.

URBANISM EXPANDED

For many people, *urban* is a bad word that implies crime, congestion, poverty, and crowding. For them, it represents an environment that moves people away from a healthy connection with nature and the land. Its stereotype is the American ghetto, a crime-ridden concrete jungle that simultaneously destroys land, community, and human potential. The reaction to this stereotype has been a middle-class retreat into the closeted world of single-family lots and gated subdivisions in the suburbs. As a result, much of the last half century's planning has been directed toward depopulating cities whether through the satellite towns of Europe or the suburbs of America.

But, for many others, the word *urban* represents economic opportunity, culture, vitality, innovation, and community. This positive reading is now manifest in the revitalized centers of many of our historic cities. In these core areas, the public domain with its parks, walkable streets, commercial centers, arts, and institutions is once again becoming rich and vibrant, valued and desirable. There is new life in many city centers and their public places, from cafes and plazas to urban parks and museums ultimately drawing people back to the city.

In fact, since 2000, many of our major cities have increased their share of new home construction while their region's suburbs have declined. For example, in 2008, Portland issued 38 percent of all the building permits within its region, compared to an average of 9 percent in the early 1990s; Denver accounted for 32 percent, up from 5 percent; and Sacramento accounted for 27 percent, up from 9 percent. There is an even stronger trend toward urban redevelopment in the largest metropolitan regions. New York City accounted for 63 percent of the building permits issued within its region. By comparison, the city averaged about 15 percent of regional building permits during the early 1990s. Similarly, Chicago now accounts for 45 percent of the building permits within its region, up from just 7 percent in the early 1990s. This represents a dramatic turnaround as cities regain their roles as centers of innovation, social mobility, artistic creativity, and economic opportunity.

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Urbanism of this caliber is desirable but, unfortunately, too often limited and very expensive. A home in the metropolitan center is, in some places, the most valuable in the region—an economic signal of just how desirable good urban places can be. In such cities as New York, Portland, Seattle, or Washington, DC, urban residences command a premium of 40 to 200 percent per square foot over their suburban alternative. Meanwhile, in our ghettos and first-ring suburbs, the working poor—and now even the middle class—are suffering and struggling. Urbanism is again proving its value: but if in limited supply, it soon can become too valuable.

At the same time, the bread-and-butter subdivisions at the metropolitan fringe experienced the greatest fall in value during the 2008 housing bust. Their physical environments along with their economic opportunities, cost of transportation, and social structures are becoming more and more stressed. Many economic and social factors are at work in this equation, but certainly a better form of urbanism is one necessary component of the renewal we need. But first, a clear definition of urbanism is needed.

Much confusion surrounds the differences between suburbs, sprawl, and what I mean by urbanism. Suburbs are not always sprawl and can be urban in many ways. Sprawl is a specific land use pattern of single-use zones, typically made up of subdivisions, office parks, and shopping centers strung together by arterials and highways. It is a landscape based on the automobile. We all know it when we see it; nevertheless, much of the debate about sprawl and urbanism is rife with misrepresentations.

For example, sprawl is typically described as discontinuous developments that wastefully hopscotch across the landscape. But healthy forms of suburban growth can also be discontinuous, as villages and towns with greenbelt separations demonstrate. Suburbs are criticized for their low densities, as if we should abolish single family homes and yards, but many great urban places integrate a full range of densities, from large-lot mansions and single-family homes to bungalows and town homes. The classic streetcar suburbs of the turn of the twentieth century were not sprawl—they were walkable, diverse in use, transit oriented and compact—but they were relatively low density and outside the city center, in a word "suburban." Conversely, urban renewal programs transformed decaying urban districts into denser versions of suburban sprawl, substituting superblocks and arterials for walkable streets and single-income projects for complex, mixed-use neighborhoods.

It is the quality of the place that is most significant in sprawl: its relentless parking lots and oversized roads, uniform tracks of houses, isolated office parks, strip commercial areas, and, above all, its near total dependence on the car. To be against sprawl is not to be against suburbs or small towns. All suburbs are not sprawl, and unfortunately, not all sprawl is suburban.

Traditional urbanism has three essential qualities: (1) a diverse population and range of activities, (2) a rich array of public spaces and institutions, and (3) human scale in its buildings, streets, and neighborhoods. Most of our built environment, from city to suburb, manifested these traits prior to World War II. Now, most suburbs succeed in contradicting each trait; public space is withering for lack of investment, people and activities are segregated by simplistic zoning, and human scale is sacrificed to a ubiquitous accommodation of the car.

None of these urban design principles are new. Jane Jacobs postulated a similar definition of urbanism in her landmark 1961 work *The Death and Life of Great American Cities*. The difference here is that urban issues are also being considered in the context of climate change and environmental protection. In fact, one can arrive at the same design conclusions from the criteria of conservation, environmental quality, and energy efficiency that Jacobs located largely by social and cultural needs. By investigating the technologies and formal systems scaled for limited resources, climate change concerns add a new and critical element to Jacobs' rationale. If traditional urbanism and sustainable development can truly reduce our dependence on foreign oil, limit pollution and greenhouse gases, and create socially robust places, they not only will become desirable but will be inevitable.

To Jacobs' three traditional urban values of civic space, human scale, and diversity, the current environmental imperative adds two more: conservation and regionalism. Although the traditional city was by necessity energy and resource efficient, it commonly showed a destructive disregard for nature and habitat that would be inappropriate today. Bays were filled, wetlands drained, streams and rivers diverted, and key habitat destroyed. A green form of urbanism should protect those critical environmental assets while reducing overall resource demands.

Indeed, the simple attributes of urbanism are typically a more cost efficient environmental strategy than many renewable technologies. For example, in many climates, a party wall is more cost effective than a solar collector in reducing a home's heating needs. Well-placed windows and high ceilings offer better lighting than efficient fluorescents in the office. A walk or a bike ride is certainly less expensive and less carbon intensive than a hybrid car even at 50 MPG. A convenient transit line is a better investment than a "smart" highway system. A small cogenerating electrical plant that reuses its waste heat locally could save more carbon per dollar invested than a distant wind farm. A combination of urbanism and green technology will be necessary, but the efficiency of urbanism should precede the costs of alternate technologies. As Amory Lovins of the Rocky Mountain Institute famously advocates, a "nega-watt" of conservation is always more cost effective than a watt of new energy, renewable or not. Urban living in its many forms turns out to be the best type of conservation.

In addition, the idea of "conservation" in urban design applies to more than energy, carbon, and the environment: it also implies preserving and repairing culture and history as well as ecosystems and resources. Conserving historic buildings, institutions, neighborhoods, and cultures is as essential to a vital, living urbanism as is preserving its ecological foundations.

Regionalism sets city and community into the contemporary reality of our expanding metropolis. At this point in history, most of our key economic, social, and environmental networks extend well beyond individual neighborhoods, jurisdictions, or even cities. Our cultural identity, open space resources, transportation networks, social links, and economic opportunities all function at a regional scale—as do many of our most challenging problems, including crime, pollution, and congestion. Major public facilities, such as sports venues, universities, airports, and cultural institutions, shape the social geography of our regions as well as extend our local lives.

We all now lead regional lives, and our metropolitan form and governance need to reflect that new reality. In fact, urbanism can thrive only within the construct of a healthy regional structure. The tradition of urbanism must be extended to an interconnected and interdependent regional network of places, creating polycentric regions rather than a metropolis dominated by the old city/suburb schism.

This last point is critical to understanding urbanism and the climate change challenge. City life is not the only environmental option; a regional solution can offer a range of lifestyles and community types without compromising our ecology. A well designed region, when combined with aggressive conservation strategies, extensive transit systems, and new green technologies, can offer many types of sustainable lifestyles. New York City may have among the smallest carbon footprint per capita, but to solve the climate change crisis we do not all have to live in the city.

Identifying an appropriate balance among technology, urban design, and regional systems in confronting climate change is now the critical challenge. As a greater percentage of the world's population increases its wealth, the definition of prosperity will become critical. If progress translates into the old American suburban lifestyle, we are all in trouble. If China and India adopt our development patterns—auto-oriented, low-density lifestyles or even a high-rise, high-density version of the same—we will truly need breakthrough technologies to accommodate the demands. If they develop an enlightened and indigenous form of urbanism, we all will have the opportunity to address climate change in a less heroic and more cost effective way.

In fact, many developing countries are fast approaching a tipping point of urbanism. As auto ownership grows, the infrastructure to support it expands. Slowly at first, then in a landslide, the logic of surface parking lots, low-density development, freeways, and malls becomes irresistible. As cars make remote destinations viable, the historic logic of density and urbanism erodes and the economics of single-use, low density suburbs grows. The built environment shifts to focus on auto mobility in ways that are hard to reverse—and with this shift urban culture dies. Traditional landscapes and neighborhoods are demolished at astonishing rates to make way for what is now seen as modern. Certainly, we cannot romanticize or literally replicate the complex historic urban fabric of, say, the Hutong in Beijing, but we can learn from it.

At the center of energy and carbon problems in the United States (and in many developing countries in the not-too-distant future) is transportation. It represents almost a third of current U.S. GHG emissions and is the fastest-growing segment. As industry becomes more efficient and jobs continue to shift toward an information economy, transportation becomes a more dominant issue.

It seems obvious that the more we spread out, the more we must drive. But the numbers are still startling.

From 1980 to 2005, average miles driven per person increased by 50 percent in the United States, a change that can be linked to the nearly 20 percent increase in land consumed per person over roughly the same period. By comparison, Portland, Oregon, with its regional focus on transit and walkable neighborhoods, has seen a reduction in vehicle miles traveled per capita since the mid-1990s. At the same time that it reduced auto dependence, the Portland region has preserved valuable farmlands and provided a widening range of housing options. Short of such regional efforts, even a doubling of auto efficiency will not keep up with the typical growth in sprawl-induced travel. We cannot solve the carbon emission problem without changing our travel behavior, and to do that an alternative to our auto-dominated communities is essential.

The good news is that truly great urban places also happen to be the most environmentally benign form of human settlement and are at the heart of a green future. Cities and urban places produce the smallest carbon footprint on a per capita basis. New Yorkers, for example, emit just a third of the GHG of the average American. In addition, it is generally accepted that population growth in developing countries drops as a rural population urbanizes. Urbanism therefore leads to fewer people consuming fewer resources and emitting less GHG at a global scale. Urbanism is a climate change antibiotic and our most affordable solution to foreign oil dependence. Urbanism is, in fact, our single most potent weapon against climate change, rising energy costs, and environmental degradation.

Yet our towns, cities, and regions cannot be shaped around a single issue like climate change or peak oil, no matter how critical they may be. Urban design is part art, social science, political theory, engineering, geography, and economics. I believe it is necessarily all of the above—urban design cannot and should not be reduced to any single metric. In the end, great urban places are qualitative: they are ultimately defined by the coherence of their public places, the diversity of their population, and the opportunity they create for our collective aspirations. We will never treasure our cities and towns just because they are low carbon, energy efficient, or even economically abundant; we will treasure them only when we come to love them as places—as vessels of our cultural identities, stages for our social interaction, and landscapes for our personal narratives. But that does not mean that they should not also play a critical role in the climate change challenge.

URBANISM AND GREEN TECHNOLOGY

. . . [W]e need to find the simple, elegant solutions that are based on conservation before we introduce complex technology, even if it is green.

We need to focus, ironically, on ends, not means. For example, in passive solar buildings, focusing on the end goal (thermal comfort) rather than the means (heating air) changed the design approach dramatically. It turns out that human comfort has more to do with surrounding surface temperatures than with air temperature in a building, so massive walls that absorb and store the sun's gentle heat also provide a more comfortable environment without all the hot air. Or, if lighting is the goal, electricity and bulbs are just one potential means; a building that welcomes daylight is the simple, elegant solution—even better than a complex system of wind farms generating green electrons for efficient fixtures. Likewise, the goal of transportation is access, not movement or mobility per se; movement is a means, not the end. So, bringing destinations closer together is a simpler, more elegant solution than assembling a new fleet of electric cars and the acres of solar collectors needed to power them. Call it "passive urbanism."

Once demands are reduced by passive urbanism, the next step is to add technology. Green urbanism is what you get when you combine the best of traditional urbanism with renewable energy sources, advanced conservation techniques, new green technologies, and integrated services and utilities. All the inherent benefits of urbanism can be amplified by a new generation of ecological design, smart grids, climate-responsive buildings, low-carbon or electric cars, and next-generation transit systems.

These technologies function in differing ways at differing scales. There are three scales of such green technology: building, community, and utility. Building-scale technologies are ecumenical; they can be applied in any form of development, traditional urban or auto-oriented sprawl. Obviously, better building insulation, weatherization, and efficient appliances can be used in single-family subdivisions as well as in urban townhomes. So, too, can solar domestic hot water systems or photovoltaic cells. Efficient light bulbs make sense in any location, as do efficient appliances. While bigger, less efficient buildings will cost more to green, such retrofits and new building standards are the starting point for any sustainable future—but not the final solution.

At the other end of the spectrum are the centralized utility-scale systems. Shifting to massive renewable sources in remote locations will carry the burden of building equally massive distribution facilities. Such a "smart grid," while essential to moving large quantities of power to our cities from distant natural resource areas (wind, sun, geothermal), has a high capital cost and reduces efficiency because of transmission line losses. These expenses are in addition to costs that are already consistently higher than those of conservation. Also, large-scale solar and wind operations can create big environmental footprints, as large tracts of virgin land are developed.

What are the real needs for large utility-scale renewable energy sources? It depends on the type of communities we plan and how we build them. If we add the travel demand of an average single-family home in the United States to the energy needed to heat, cool, and power the home, the total is just under 400 million Btu (British thermal units) per year (this includes the source energy typically left out of these calculations: the embodied energy of cars, the energy to produce the gasoline, and the wasted energy to produce the home's electricity). Assume for argument that weatherization and greening this home can reduce building energy consumption by 30 percent and that the family buys new cars with 50 percent better mileage. The result is a 32 percent overall energy reduction—not bad for "green sprawl." In contrast, a typical townhome located in a walkable neighborhood (not necessarily downtown but near transit) without any solar panels or hybrid cars consumes 38 percent less energy than such a suburban single-family home. Traditional urbanism, even without green technology, is better than green sprawl.

Now add more building conservation measures, green technology, and better transit systems to the townhouse, and you get close to the results we will need in 2050. If you move to a green townhome in a transit village, you will be consuming 58 percent less energy than on a large lot in the suburbs. If you move to a green condo in the city, you will be saving 73 percent when compared to the average single-family home in a distant suburb.

The implications of this for our power grid are massive. If more families lived this way—say just a quarter moved from single-family lots to green townhomes—the generating capacity required for buildings in the nation would be reduced by over 25,000 megawatts per year, eliminating the need for 50 new 500-megawatt plants.

At $1.3 million per megawatt of installed capacity, that is more than $32 billion of avoided capital cost for new power plants per year. The reduced fuel costs and environmental impacts are additional benefits.

The same is true for auto use. For example, satisfying California's need for more driving in a "Trend" future would result in around 183 billion additional auto miles per year in 2050 when compared to the more urban alternative. Some believe that if we shifted to electric cars running on green electrons, the carbon problem could be solved, However, producing that many green electrons has a hidden hurdle: it would take 50,000 acres of high-efficiency solar thermal plants, 130,000 acres of photovoltaic panels, or 860,000 acres of wind farms (nearly thirty times the land area of San Francisco) to power such a transportation system. This would present a giant environmental footprint no matter where it was placed. Ironically, the biggest barrier to such a green, if not urban, solution may be environmentalists themselves, protesting lost desert landscapes or resisting impacts on bird populations by wind turbines (or even objecting to seeing the turbines on the horizon).

At the middle of the three scales, urbanism offers a better framework for more distributed community-scale energy systems. In fact, there are important community-scale systems that can function only within an urban framework. One of the most significant of these technologies is the decentralized cogeneration electric power plant (called combined heat and power, or CHP). Such small-scale power plants can be coupled with district heating and cooling systems to capture and use the generator's waste heat in local buildings and industry. Currently, for every watt of energy delivered to a home, two thirds is lost as waste heat up the smokestack and in transmission lines. Local cogeneration plants coupled with district heating and cooling systems can largely eliminate these inefficiencies. The waste heat is captured and reused, while the transmission losses are greatly reduced. Because of this, it is estimated that cogeneration systems operate at around 90 percent efficiencies whereas standard power plants average only 40 percent.

Married to urban environments, cogeneration offers a cheap, time-tested alternative—one that has been employed by college campuses and European new towns for decades. There, small power plants arc placed close to dense neighborhoods and commercial centers, distributing waste heat underground to each building for hot water, cooling, and heating. These plants can burn almost any form of renewable biomass, eliminating the energy-intensive process of converting valuable crops into biofuels or finding mechanisms to transform grass to gas. More interesting are a new generation of "waste to energy" technologies that not only produce green electricity and heat but also avoid the massive landfills and trucking costs of typical garbage systems.

Typically, cogeneration systems are found in commercial applications where waste heat is used in an industrial process and the power generation balances with the electrical demand. It is estimated that in the industrial sector alone, "the potential for CHP generation is equivalent to the output of 40 percent of the coal fired generating plants in the US." Utilizing similar systems in urban districts would add dramatically to this potential.

Sacramento built such a system in its downtown in the 1970s that burned "gasified" dead wood created by a Sierra Mountain beetle infestation—a net zero carbon system because it used only biomass. In addition, it had twice the efficiency of a remote plant because its waste heat was used to run heaters and chillers for all the state office buildings in the district. But to be effective, such systems are dependent on urban densities and a balanced mix of uses. Sprawl is not a candidate for district heating and cooling systems, as the costs of moving the waste heat to scattered buildings are too high. However, mixed-use urban neighborhoods could top off their energy needs with cogeneration in ways that greatly reduce costs and environmental impacts—easily creating zero net energy communities.

Water and waste systems also benefit from a community-scale approach. Sewer systems can take effluent and biologically recycle it into potable or irrigation water, usable biomass, and methane for cooking. Water demands can be offset by such graywater recycling systems, drought-tolerant landscaping, and indigenous plantings. Stormwater detention and treatment can be decentralized to community-scaled parks and integrated as landscape features. Rather than channelizing streams and rivers, setbacks can allow habitat to coexist with flood protection and trails. As with energy systems, community-scaled water and waste systems can be ecologically integrated in ways that save costs, save carbon, and enhance livability.

TRANSIT: THE GREENEST TECHNOLOGY

Of course, the most important community-scale system dependent on urbanism is transit. It has long been known that density and transit ridership are linked, but it goes much deeper than that. The key to viable transit systems is not just density but walkability and mixed use—true urban places. If people cannot walk the quarter mile to or from a station, chances are they will not use the transit. Conversely, if they can easily run errands and coordinate trips on the way to or from a station, they are more likely to use transit. European data show that the percentage of walk or bike trips always exceeds that of transit trips—often by more than two to one; In fact, walking by itself constitutes 30 percent of all trips in Great Britain (versus 9 percent transit), and in Sweden walk/bike trips are 34 percent of the total (versus 11 percent transit). Transit supports and extends the pedestrian environment; transit is pedestrian dependent, not the other way around. The primary alternative to the car and all of its environmental costs is the pedestrian environment and the walkable urbanism that supports transit.

A good transit system has many layers, from local buses to bus rapid transit and streetcars, from light rail to subways and commuter trains. They all feed into and reinforce one another, and they all depend on walkable urbanism at the origin and destination. The quality of the interface from walking to transit, and from one form of transit to the other, is central to displacing car trips and is the greenest technology that urbanism provides.

The relationship among transit, urbanism, travel behavior, and carbon emissions is complex but can be summarized with one key quantifiable metric, vehicle miles traveled—effectively, the amount we drive. VMT is determined by the number and distance of trips we take, and our "mode split"—the percentage of trips taken by various transportation modes such as walk, bike, car, carpool, or transit. Each household, depending on its location, income, and size, has an average VMT per year, which when combined with various auto technologies will generate its travel carbon footprint.

Many factors affect VMT, and there are many complex models that simulate the travel behavior behind it. For example, the modal split among auto, walk/bike, and transit is affected by location and level of transit service as well as how pedestrian friendly the streets are; the average length of each type of trip is affected by land use patterns and how closely destinations are located; the number of trips per day is affected by household size; and auto ownership rate is affected by household income and size . . . The most significant variables in all this are the walking and transit opportunities of urbanism, a compact development form, and land use patterns that bring destinations closer together.

The power of place over travel behavior is demonstrated by mapping VMT per household across a region. While averages always lead us to stereotypes, different environments across any region reveal dramatically different travel behaviors. For example, in the San Francisco Bay Area, a typical household in the Russian Hill neighborhood of San Francisco has an average VMT of 7,300 miles a year. This neighborhood averages only three stories but is dense by suburban standards; has a rich mix of shops, restaurants, and services within walking distance; and is a short transit ride from downtown. Its walk score (an algorithm that awards points based on the distance to the closest amenity in several categories) is 98 out of 100—as good as it gets.

The Rockridge neighborhood in Oakland was created as a streetcar suburb back in the prewar days of the Key Route Trolley system, which connected most of the Bay Area until 1948. It is filled largely with bungalow and small-lot single-family homes but has small apartment buildings at corners and a wonderful mixed-use main street along with a BART (Bay Area Rapid Transit) train station at its center. The average household there drives about 12,200 miles a year and has a walk score of 74.

Out in San Ramon, a low-density East Bay suburb without good transit connections, development patterns fit the standard sprawl paradigm, with isolated single-family subdivisions, strip commercial arterials, malls, and office parks. VMT for the average home there is around 30,000 miles a year, and the walk score is 46.

So there is a four-to-one range in travel behavior over three neighborhoods in one region. They differ in density, mix of uses, walkability, proximity to job centers, and level of transit service. The density in Russian Hill is 62 units per acre, but home values are $555 per square foot. In Rockridge, the density averages 15 units per acre and values are $420 per square foot. Finally, in San Ramon, considered a very high end suburban community, the average density is 3.4 units per acre and the value averages just $320 per square foot. The market itself is telling us that walkable places have value and, as a bonus, can reduce our carbon emissions and oil dependence. So desirable is the walkable neighborhood that a 2009 study found that in cities like San Francisco and Chicago, moving from a household with a city's median walkability to one at the 15th percentile would increase the unit's value by over $30,000. The challenge, of course, is to create walkable places as authentic and beautiful as Russian Hill and Rockridge that are affordable.

The point is that all of these community-scale systems—whether power, water, waste, or transit—need urbanism to be effective. Urbanism is essential for the viability of community cogeneration systems and the savings they provide in energy consumption.

Denser, mixed-use development can provide the open space, community parks, and riparian setbacks needed by ecological water and waste recycling systems, and, of course, transit depends on urbanism for its fundamental viability.

These community-scale systems built around urbanism are not intended to replace the emissions reductions of efficient industrial processes, renewables in our utility portfolios, or better fuel standards for our cars. It is just that those supply-side strategies alone will not take us far enough quickly enough—and they come at a large cost premium. The combination of transit-served urbanism and green technology at the community scale is essential to complete the picture.

All of this discussion boils down to some simple choices in community building. One alternative simply extends our current land use patterns, architectural types, everyday aesthetics, and civic habits. As one example of this, imagine a room with a low-hung ceiling, sealed windows, and fluorescent lights; within a building with a mirror glass skin, set behind a parking lot off a six-lane arterial; in a zone of commercial development making up part of a suburb of subdivisions, shopping centers, and office parks connected by a freeway to a metropolis of decaying inner-city neighborhoods, struggling first-ring suburbs, exclusive suburban enclaves, failing school systems, and underfunded civic programs. This would seem like a biased contrivance if it were not so commonplace.

The other choice involves a quality of place making we seem to have lost touch with. It could be described as a room with high ceilings filled with natural light and breezes; in a building wrapping a courtyard and lining a street; in a neighborhood with tree-lined avenues, village greens, and local shops; making up a part of a city filled with streetcars, public squares, parks. and cultural districts; providing the focus of a metropolis with a constellation of many varied towns and cities connected by transit, growing economic networks, cultural institutions, and social opportunity, This also may seem like a biased contrivance, but it has been realized in some significant U.S. metro areas.

In both models, each layer is interdependent and connected by deep-rooted economic, policy, and social systems. Each is a complex that cannot easily exist piece by piece but nests layer by layer into a self-reinforcing "whole system." Certainly, the future will be a mix of these two extremes, but the question is: in what proportions?

"Charter of the New Urbanism"

Congress for the New Urbanism

EDITORS' INTRODUCTION

The sustainability principles espoused by the Brundtland Commission (p. 404) helped to give weight and authority to urban environmentalist organizations worldwide, none more so than an innovative planning and design movement called the New Urbanism. The Chicago-based Congress for a New Urbanism was officially established in 1993, but the immediate origin of the movement was a meeting at the Awahnee Hotel in Yosemite Valley, California, in 1991. There, an extraordinary collective of visionary architects and designers – among them, Peter Calthorpe, Andres Duany, Elizabeth Plater-Zyberk, Michael Corbett, Stafanos Polyzoides, Daniel Solomon, and Elizabeth Moule – met with a number of California policy-makers to promulgate the Awahnee Principles for future urban development along ecologically sound lines. Many of those Principles became elements of the "Charter of the New Urbanism." The movement made swift gains throughout the 1990s – becoming a favored model of the US Department of Housing and Urban Development during the Clinton administration – and New Urbanist projects were built throughout the United States and Canada. In 2003, an allied Council for European Urbanism was established in the UK, actively encouraged by HRH Charles, Prince of Wales, and the New Urbanism spread worldwide with projects in France, Portugal, Sweden, Italy, Belgium, the Netherlands, Australia, New Zealand, South Africa, and China.

The heart of the Charter of the New Urbanism consists of 27 principles – nine in each of three broad categories – preceded by a kind of preamble that establishes the visionary, almost utopian, goals of the movement. The preamble begins by asserting that all of today's urban ills – inner-city decay, suburban sprawl, the deterioration of agricultural and wilderness lands, even race- and class-based segregation – are parts of "one interrelated community-building challenge." It goes on to call for the "restoration of existing urban centers" and the transformation of "sprawling suburbs into communities of real neighborhoods and diverse districts." New urbanism recognizes "that physical solutions by themselves will not solve social and economic problems," but it insists that "a coherent and supportive physical framework" is a necessary, if not sufficient, precondition for urban progress and that such progress must be achieved "through citizen-based participatory planning and design."

The 27 principles of the Charter address contemporary urban planning issues at a much finer level of detail, beginning with an examination of cities and towns at the metropolitan scale. "The metropolitan region," it states, is defined by natural topography and represents "a fundamental economic unit of the contemporary world." These metropolitan-scale principles go on to call for urban growth boundaries that do not "blur or eradicate the edges of the metropolis," intensive "infill development" within existing cities, region-wide revenue sharing, and a wide range of transportation options that "maximize access and mobility . . . while reducing dependence upon the automobile."

The next set of principles examines the needs of "the neighborhood, the district, and the corridor." The Charter calls for neighborhoods that are "compact, pedestrian-friendly, and mixed-use" so that "many activities of daily living" can be within walking distance. In addition, neighborhoods should contain local shopping districts (not distant malls), parks, and community schools. Finally, another nine principles look at "the block, the street, and the building," calling for an architecture that "transcends style," that grows from "local climate, topography, history, and building practice," and that creates environments characterized by safety, accessibility, and openness.

The New Urbanism is not without its critics. Some have dismissed it as a "New Suburbanism" that addresses the issues of the young middle-class – double-income, no kids families called DINKs – but that has no relevance for low-income inner-city neighborhoods or even the loft-living districts of the tech workers of the Millennial Generation. Others feel that the "new traditionalism" tendencies of many New Urbanist developments feel artificial and too carefully, too strictly planned. And one critic even claimed that the emphasis on openness and accessibility leads to "crime-friendly neighborhoods." But for all this, the New Urbanism has proven to be a long-lived and ever-evolving movement.

Unlike most of the twentieth-century planning movements, the New Urbanism is not tied to the ambitions and pretensions of a single individual. Rather like the Garden City movement that Ebenezer Howard pioneered but did not monopolize, the New Urbanism has attracted a large number of practitioners, and the movement has various wings and branches that continually question and inform the movement's mainstream. For example, although the Charter of the New Urbanism calls for a reasonable mix of transit options, including the private automobile, one somewhat alarmist video documentary of 2004 is titled *The End of Suburbia: Oil Depletion and the Collapse of the American Dream*. Yet another video, *New Urban Cowboy: Toward a New Pedestrianism* of 2008, appears to be a publicity vehicle for the producer's campaign for the Florida governorship! For a more sober critique, see David Harvey, "The New Urbanism and the Communitarian Trap," *Harvard Design Magazine*, 1 (1997), pp. 68–69.

The literature on the New Urbanism is as rich and varied as the movement itself. Peter Katz, *The New Urbanism: Towards an Architecture of Community* (New York: McGraw-Hill, 1994) and Doug Kelbaugh, *Common Place: Toward Neighborhood and Regional Design* (Seattle: University of Washington Press, 1997) provide overviews of designs by Calthorpe and other New Urbanists. Kenneth B. Hall and Gerald A. Porterfield, *Community by Design: New Urbanism for Suburbs and Small Communities* (New York: McGraw-Hill Professional, 2001) and E. Talen, *New Urbanism and American Planning: The Conflict of Cultures* (London and New York: Routledge, 2005) offer detailed analyses of the New Urbanist movement. The 69-page *New Urbanism: Peter Calthorpe vs. Lars Lerup: Michigan Debates on Urbanism* (Ann Arbor: University of Michigan Press, 2005) is a lively and scintillating exchange of views with an afterword by editor Robert Fishman. See also James Howard Kunstler's *The Geography of Nowhere* (New York: Simon & Schuster, 1993) and *Home from Nowhere: Remaking Our Everyday World for the Twenty-first Century* (New York: Simon & Schuster, 1998) for a popular account of Calthorpe and other New Urbanists' work. Also of interest are John Dutton, *New American Urbanism: Re-forming the Suburban Metropolis* (Milan: Skira, 2001), Todd W. Bressi (ed.), *The Seaside Debates: A Critique of the New Urbanism* (New York: Rizzoli, 2002), and Gabriele Tagliaventi, *New Urbanism* (Florence: Alinea, 2002). The best place to begin any research on the New Urbanist movement is Michael Leccese and Kathleen McCormick (eds.), *Charter of the New Urbanism* (New York: McGraw-Hill, 1999).

THE CONGRESS FOR THE NEW URBANISM views disinvestment in central cities, the spread of placeless sprawl, increasing separation by race and income, environmental deterioration, loss of agricultural lands and wilderness, and the erosion of society's built heritage as one interrelated community-building challenge.

WE STAND for the restoration of existing urban centers and towns within coherent metropolitan regions, the reconfiguration of sprawling suburbs into communities of real neighborhood and diverse districts, the conservation of natural environments, and the preservation of our built legacy.

WE RECOGNIZE that physical solutions by themselves will not solve social and economic problems, but neither can economic vitality, community stability, and environmental health be sustained without a coherent and supportive physical framework.

WE ADVOCATE the restructuring of public policy and development practices to support the following principles: neighborhoods should be diverse in use and population; communities should be designed for the pedestrian and transit as well as the car; cities and towns should be shaped by physically defined and universally accessible public spaces and community institutions; urban places should be framed by architecture and landscape design that celebrate local history, climate, ecology, and building practice.

WE REPRESENT a broad-based citizenry, composed of public and private sector leaders, community activists, and multidisciplinary professionals. We are committed to reestablishing the relationship between the art of building and the making of community, through citizen-based participatory planning and design.

WE DEDICATE ourselves to reclaiming our homes, blocks, streets, parks, neighborhoods, districts, towns, cities, regions, and environment.

We assert the following principles to *guide public policy, development practice, urban planning, and design:*

The region: Metropolis, city, and town

1. Metropolitan regions are finite places with geographic boundaries derived from topography, watersheds, coastlines, farmlands, regional parks, and river basins. The metropolis is made of multiple centers that are cities, towns, and villages, each with its own identifiable center and edges.
2. The metropolitan region is a fundamental economic unit of the contemporary world. Governmental cooperation, public policy, physical planning, and economic strategies must reflect this new reality.
3. The metropolis has a necessary and fragile relationship to its agrarian hinterland and natural landscapes. The relationship is environmental, economic, and cultural. Farmland and nature are as important to the metropolis as the garden is to the house.
4. Development patterns should not blur or eradicate the edges of the metropolis. Infill development within existing urban areas conserves environmental resources, economic investment, and social fabric, while reclaiming marginal and abandoned areas. Metropolitan regions should develop strategies to encourage such infill development over peripheral expansion.
5. Where appropriate, new development contiguous to urban boundaries should be organized as neighborhoods and districts, and be integrated with the existing urban pattern. Noncontiguous development should be organized as towns and villages with their own urban edges, and planned for a jobs/housing balance, not as bedroom suburbs.
6. The development and redevelopment of towns and cities should respect historical patterns, precedents and boundaries.
7. Cities and towns should bring into proximity a broad spectrum of public and private uses to support a regional economy that benefits people of all incomes. Affordable housing should be distributed throughout the region to match job opportunities and to avoid concentrations of poverty.
8. The physical organization of the region should be supported by a framework of transportation alternatives. Transit, pedestrian, and bicycle systems should maximize access and mobility throughout the region while reducing dependence upon the automobile.
9. Revenues and resources can be shared more cooperatively among the municipalities and centers within regions to avoid destructive competition for tax base and to promote rational coordination of transportation, recreation, public services, housing, and community institutions.

The neighborhood, the district, and the corridor

1. The neighborhood, the district, and the corridor are the essential elements of development and redevelopment in the metropolis. They form identifiable areas that encourage citizens to take responsibility for their maintenance and evolution.
2. Neighborhoods should be compact, pedestrian-friendly, and mixed-use. Districts generally emphasize a special single use, and should follow the principles of neighborhood design when possible. Corridors are regional connectors of neighborhoods and districts; they range from boulevards and rail lines to rivers and parkways.
3. Many activities of daily living should occur within walking distance, allowing independence to those who do not drive, especially the elderly and the young. Interconnected networks of streets should be designed to encourage walking, reduce the number and length of automobile trips, and conserve energy.
4. Within neighborhoods, a broad range of housing types and price levels can bring people of diverse ages, races, and incomes into daily interaction, strengthening the personal and civic bonds essential to an authentic community.

5. Transit corridors, when properly planned and coordinated, can help organize metropolitan structure and revitalize urban centers. In contrast, highway corridors should not displace investment from existing centers.
6. Appropriate building densities and land uses should be within walking distance of transit stops, permitting public transit to become a viable alternative to the automobile.
7. Concentrations of civic, institutional, and commercial activity should be embedded in neighborhoods and districts, not isolated in remote, single-use complexes. Schools should be sized and located to enable children to walk or bicycle to them.
8. The economic health and harmonious evolution of neighborhoods, districts, and corridors can be improved through graphic urban design codes that serve as predictable guides for change.
9. A range of parks, from tot-lots and village greens to ball fields and community gardens, should be distributed within neighborhoods. Conservation areas and open lands should be used to define and connect different neighborhoods and districts.

The block, the street, and the building

1. A primary task of all urban architecture and landscape design is the physical definition of streets and public spaces as places of shared use.
2. Individual architectural projects should be seamlessly linked to their surroundings. This issue transcends style.
3. The revitalization of urban places depends on safety and security. The design of streets and buildings should reinforce safe environments, but not at the expense of accessibility and openness.
4. In the contemporary metropolis, development must adequately accommodate automobiles. It should do so in ways that respect the pedestrian and the form of public space.
5. Streets and squares should be safe, comfortable, and interesting to the pedestrian. Properly configured, they encourage walking and enable neighbors to know each other and protect their communities.
6. Architecture and landscape design should grow from local climate, topography, history, and building practice.
7. Civic buildings and public gathering places require important sites to reinforce community identity and the culture of democracy. They deserve distinctive form, because their role is different from that of other buildings and places that constitute the fabric of the city.
8. All buildings should provide their inhabitants with a clear sense of location, weather and time. Natural methods of heating and cooling can be more resource-efficient than mechanical systems.
9. Preservation and renewal of historic buildings, districts, and landscapes affirm the continuity and evolution of urban society.